DNSC 6330 Individual Homework 5

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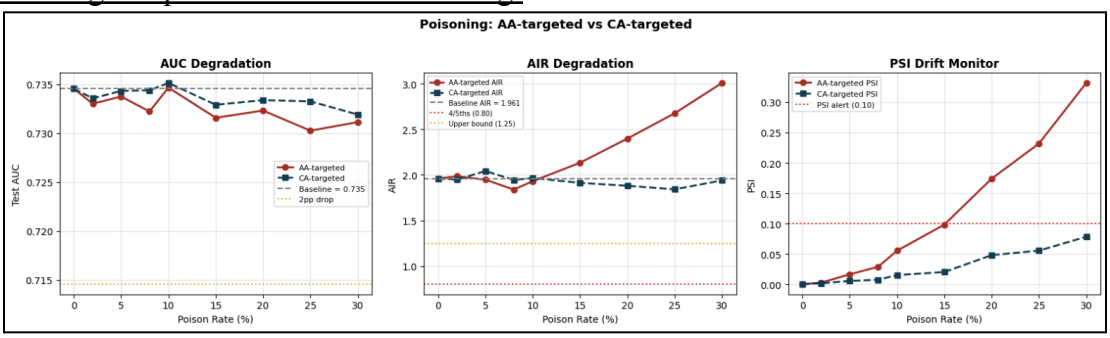
(1) PGD Evasion Audit Results:

For logistic regression, the attack works fast. At just ε = 0.25, the FPR for African-American defendants jumps from 28% to 56.9%, while Caucasian defendants go from 14% to 37%. As ε keeps increasing, both groups eventually get flagged at nearly 100%, and the AIR drops from 1.96 all the way down to 1.0, meaning the model is now wrong about almost everyone equally. The AIR never actually crosses below 0.80, but only because the attack destroys the model for both groups at the same time rather than being selectively worse for one. For GBT, the attack does essentially nothing. FPR stays fixed at 0.317 for African-Americans and 0.178 for Caucasians across every epsilon value tested. This makes sense intuitively, gradient boosted trees make discrete, step-wise decisions, so small continuous adjustments can't push predictions across a decision boundary the way they can for a smooth model like logistic regression.

In terms of choosing a model for deployment, it's not a super clear answer from these results. GBT is clearly more robust to this type of evasion attack, but look at its baseline: AIR of 1.782 means African-American defendants are already being falsely flagged at nearly twice the rate of Caucasian defendants before any attack even happens. Robustness to adversarial inputs doesn't fix a pre-existing fairness problem. LR has a better baseline AIR but falls apart under attack. Adversarial robustness and baseline fairness both matter, and a model that passes one test can easily fail the other.

(2) Poisoning Loop with Fairness Monitoring:

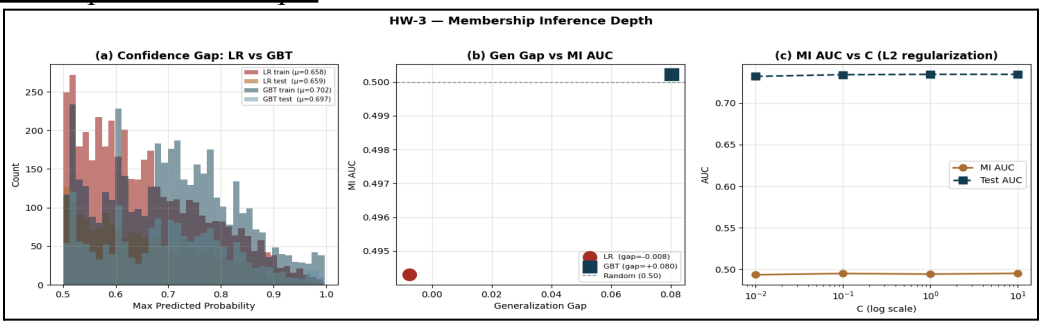


(a) The plots show both variants on the same axes. AUC stays nearly flat for both attacks across all poison rates, hovering around 0.733–0.735 throughout. The AIR tells a different story, the AA-targeted attack keeps AIR above 1.25 the entire time (starting at 1.961 and staying above), while the CA-targeted attack actually pushes AIR in the opposite direction, eventually dropping it toward 1.0 as more Caucasian high-risk records get flipped to low-risk.

(b) Since baseline AIR is already 1.961 before any poisoning, both attacks start outside the [0.80, 1.25] window from the very first poison rate. Combined with AUC staying within 2pp throughout, both technically satisfy the stealth condition across the entire 0–30% range. For the AA-targeted attack, between 5–25% poison, delta_AUC sits at -0.001 to -0.003 while AIR is quietly climbing toward 2.4 and beyond. The attack is doing real damage to fairness the whole time with nothing in the accuracy numbers to flag it. For the CA-targeted attack, the stealth condition is satisfied mostly because the attack doesn't move AIR much to begin with. Flipping Caucasian labels shifts both group FPRs in similar proportions, so the ratio stays stable. It's stealthy not because it's cleverly designed to evade detection, but because it isn't doing much in the first place.

(c) The AA-targeted attack doesn't trigger the PSI monitor until 20% poison, meaning the first 15 percentage points of fairness degradation go completely undetected. The CA-targeted attack never triggers it at all, PSI peaks around 0.09 at 30% and stays below the threshold the entire time. PSI tells you the input distribution shifted, not that the model became more discriminatory.

(3) Membership Inference Depth



(a) For LR, train and test distributions are nearly identical with means of 0.658 and 0.659 — the model is equally uncertain about training data and unseen data. GBT shows more separation, with train (μ=0.702) shifted right and carrying a heavier tail toward higher confidence scores compared to test (μ=0.697), which lines up with its larger generalization gap.
(b) GBT has a generalization gap of +0.080 (train AUC 0.798 vs test AUC 0.718) while LR sits at -0.008, meaning LR actually generalizes slightly better on unseen data than its own training set. Despite GBT having the larger gap, both MI AUCs land at essentially chance — 0.500 and 0.494 respectively. The generalization gap didn't predict MI AUC here, a wider gap did not translate into more privacy leakage.
(c) Across C = {0.01, 0.1, 1.0, 10.0}, MI AUC barely moves , sitting flat between 0.493 and 0.495 the entire time. Test AUC does respond slightly, dropping to 0.732 at the most aggressive regularization (C = 0.01) and recovering to 0.735 at C = 1.0. So there's a real accuracy cost to heavy regularization but zero privacy benefit, since MI AUC was already at chance. L2 regularization only matters as a privacy lever when a model is actually leaking membership information, applying it here just costs predictive performance for nothing in return.

(4) Reflection

The single highest-risk finding across all three parts is the AA-targeted label-flip poisoning attack. AUC never drops more than a few tenths of a percent the entire time while AIR climbs from 1.961 to 3.010 with 30% poison. A standard monitoring setup would see nothing wrong while African-American defendants are being systematically disadvantaged at an escalating rate. It's the most dangerous result because nothing in the accuracy numbers would tell you it's happening.

- *Proactive mitigation:* fairness-constrained retraining. Rather than treating fairness as something checked after the fact, a fairness penalty gets added to the loss function during training so the optimizer has to keep disparity in check alongside minimizing prediction error. From the poisoning results, AIR starts moving at 2% poison rate while delta_AUC sits at just -0.002, so a constraint flagging AIR shifts greater than 0.10 from baseline would have caught the attack almost immediately. The disparate impact concern is that any constraint aggressively correcting for one group can shift disparity onto the other if not carefully calibrated, especially given that the baseline AIR of 1.961 reflects a pre-existing imbalance in the data itself.
- *Reactive mitigation:* direct AIR monitoring at every retraining cycle. The PSI monitor caught the AA-targeted attack only at 20% poison and missed the CA-targeted attack entirely, with PSI peaking at just 0.08ish at 30% poison, below the 0.10 threshold. Tracking AIR directly instead would have flagged the AA-targeted attack from the very first retraining cycle. A reasonable implementation would require human review any time AIR shifts more than 0.10 between retraining runs. The concern is that AIR on its own doesn't tell you why disparity is happening, it just tells you that it is. If the underlying data imbalance driving the 1.961 baseline isn't addressed separately, flagging every retraining run that drifts above the threshold treats the symptom without fixing the cause.